

From chain length to ion diffusion: Molecular insights into phosphonium-based polymerized ionic liquid membranes for energy applications

Jun Ho Park^a, Enrica Fontananova^b, Francesco Galiano^b, Alberto Figoli^b, Raffaella Mancuso^c, Bartolo Gabriele^c, Tae Kyung Lee^a, Chi Hoon Park^d, Elena Tocci^{b,*}, Sang Yong Nam^{a,*}

^a Department of Materials Engineering and Convergence Technology, Gyeongsang National University, Jinju, 52828, Republic of Korea

^b Institute on Membrane Technology–National Research Council of Italy (CNR-ITM), Via P. Bucci 17C, 87036, Rende, (CS), Italy

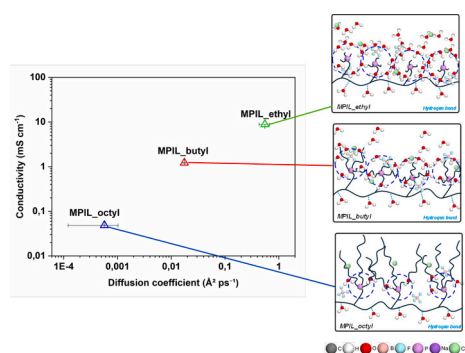
^c Laboratory of Industrial and Synthetic Organic Chemistry (LISOC), Department of Chemistry and Chemical Technologies, University of Calabria, Via P. Bucci 12/C, 87036 Rende, (CS), Italy

^d Department of Energy System Engineering, Gyeongsang National University, Jinju-si, Gyeongnam, 52849, Republic of Korea

HIGHLIGHTS

- PIL membranes with ethyl, butyl, and octyl phosphonium alkyl chains
- Experiments + MD simulations reveal chain length–ion transport link.
- Longer chains create hydrophobic domains and tune water uptake.
- Optimal length balances OH[−] transport and membrane stability.
- MD shows distinct hydration and ion conduction pathways.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Anion exchange membranes
Phosphonium-based membranes
Ion transport
Molecular dynamics simulations
Hydration
Polymer architecture

ABSTRACT

The development of high-performance anion exchange membranes (AEMs) is essential for advanced electrochemical technologies such as alkaline fuel cells and water electrolysis. Here, we investigate the influence of phosphonium side chain length on structural hydration and chloride ion transport in polymerized ionic liquid (MPIL) membranes. Atomistic molecular dynamics simulations are conducted on hydrated MPIL systems with ethyl, butyl, and octyl n-alkyl substituents, and simulation results are validated against experimental data for water uptake and ionic conductivity.

Shorter side chains (ethyl) significantly enhance water uptake (≈ 81 wt%) and promote the formation of interconnected hydrophilic channels, resulting in markedly higher Cl[−] ionic conductivity. In contrast, longer chains (octyl) restrict water accessibility and confine ion diffusion within localized hydrophilic domains, favoring ion retention tendency at the expense of transport efficiency. Intermediate chain length (butyl) yields a balanced morphology, combining moderate hydration with controllable ion mobility.

* Corresponding authors.

E-mail addresses: e.tocci@itm.cnr.it (E. Tocci), walden@gnu.ac.kr (S.Y. Nam).

<https://doi.org/10.1016/j.desal.2025.119434>

Received 13 August 2025; Received in revised form 6 September 2025; Accepted 22 September 2025

Available online 24 September 2025

0011-9164/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Quantitative analyses, including pore connectivity descriptors, ion–ion association free energies from RDF integration, and backbone–water interaction profiles, consistently confirm that steric hindrance modulates hydration shell formation, ion pairing, and channel percolation. This molecular-level insight suggests that alkyl chain engineering provides a tunable parameter for optimizing trade-offs between ion conductivity and relative mobility control in MPIL-based AEMs. The combined computational and experimental results provide practical guidelines for designing next-generation membranes for desalination, electrochemical conversion, and energy storage.

1. Introduction

The critical challenges posed by climate change have become a global priority, driving nations to implement innovative strategies and accelerate advancements in research. Central to these efforts is the development of groundbreaking technologies to address pressing environmental issues and meet the growing demand for sustainable energy solutions [1]. Among these technologies, membrane technology has emerged as a versatile and impactful solution. Its unique properties have enabled widespread application across industries such as water treatment, gas separation, and energy storage, providing effective strategies to address modern climate-related challenges [2–5]. One of the key applications of membrane technology is in ion exchange membranes (IEMs), which play an essential role in energy-related processes such as renewable energy generation, hydrogen production through water electrolysis, and energy storage in fuel cells [6–8].

Additionally, IEMs are crucial in processes like electrolysis, electrodialysis and water purification. IEMs are classified into cation exchange membranes (CEMs) [9–12], which facilitate the selective transport of cations like hydrogen ions, and anion exchange membranes (AEMs) [13–19], which enable the selective transfer and separation of hydroxide ions. While CEMs incorporate negatively charged functional groups, such as sulfonic and phosphoric acids, AEMs feature positively charged groups, including quaternary ammonium, imidazolium, and phosphonium moieties. AEMs offer distinct advantages over CEMs by operating in alkaline environments, which allow the use of non-platinum catalysts in electrolysis and fuel cell systems, significantly reducing costs [20,21]. Additionally, alkaline conditions support the efficient oxidation of alternative fuels like ammonia and alcohols, broadening the range of usable fuels [22]. However, AEMs face challenges such as lower hydroxide ion conductivity compared to CEMs and instability of functional groups in alkaline conditions [23–25]. Consequently, ongoing research is focused on developing materials with enhanced stability and performance.

One promising approach recently proposed to addressing these challenges involves polymerized ionic liquids, which integrate ionic liquid properties into macromolecular structures [26,27].

Polymerized ionic liquids exhibit unique ionic conductivity, chemical resilience, and stability in aqueous environments, making them attractive for energy applications [28–34]. Among them phosphonium-based variants are particularly noteworthy, offering superior thermal stability, lower melting points, and reduced viscosity compared to other cationic functional groups [28,35–39]. By optimizing the *n*-alkyl chain length of phosphonium functional groups and promoting phase separation in hydrophobic regions, the materials based on polymerized ionic liquids achieve remarkable chemical stability in alkaline environments. These characteristics make phosphonium-based polymerizable ionic liquid membranes (MPILs) ideal for use in AEMs for fuel cells and water electrolysis systems, where enhanced performance and durability are critical.

Despite the significant experimental progress, the molecular mechanisms governing the performance of polymerized ionic liquids-based membranes remain underexplored. Previous studies have synthesized and applied MPILs for CO₂ separation [37] and demonstrated the potential of MPIL butyl in AEMs for electrochemical applications [39].

Although various MPILs have been developed and tested [26,27], a

molecular-level understanding of how alkyl chain length influences hydration structure and ionic mobility remains incomplete. To address this, we employed atomistic molecular dynamics (MD) simulations, supported by experimental validation, to elucidate the role of side chain length on the microstructural and transport properties of hydrated MPIL membranes.

MD simulations offer unique advantages for understanding the dynamic interactions of ions, water molecules, and polymer functional groups within hydrated membrane structures. By simulating molecular interactions and dynamic processes under realistic operating conditions, MD simulations complement experimental findings and bridge the gap between molecular-level phenomena and macroscopic observations [40–52].

Here, three representative systems featuring ethyl, butyl, and octyl phosphonium substituents were modeled and compared in terms of water uptake, ionic conductivity, and hydration morphology and molecular interactions. Beyond conventional observables, we quantified ion association free energy from RDF analysis, hydration channel connectivity and polymer backbone–water interactions. Together with experimental validation, these calculations provide a comprehensive molecular-to-macroscopic picture of how alkyl chain engineering governs hydration, ion mobility, and retention tendency in MPIL membranes.

The objectives of this work are to:

- Quantify the influence of alkyl chain length on water distribution and polymer morphology.
- Elucidate the formation and continuity of hydrophilic domains relevant to ion transport.
- Establish structure–function trends that can inform the rational design of high-performance AEMs.

These findings support the design of customizable MPIL-based membranes for energy and water applications, where control over ion retention tendency and conductivity is essential.

2. Experimental

2.1. Polymer modeling

All simulations were conducted using the Forcite module in BIOVIA Materials Studio software [53]. The polymers used in the simulations were based on MPIL structures derived from prior experimental studies [37]. MPIL polymers with varying *n*-alkyl chain lengths were designed, as shown in Table 1, Table S1 and Fig. S1.

Each polymer used the same counterion, BF₄[−]. Force field parameters and atomic charges were assigned using the default settings of COMPASS III [54,55]. Parameters for the phosphonium moieties are explicitly included in COMPASS III, and partial charges were automatically generated using the bond-charge increment scheme. In all MPILs, the positive charge is primarily localized on the quaternary P⁺ atom with a minor delocalization onto the α -carbons of the substituents. No ad hoc fitting was applied. This force field, specifically parameterized for condensed-phase systems and polymers, includes refined parameters for sulfonated and quaternized functional groups commonly present in MPIL-based membranes. These enhancements make it particularly

suitable for accurately modeling hydrated ionomer membranes and related materials, capturing both structural and dynamic properties under realistic operating conditions [56,57].

The computational procedure for the polymer construction starts with building first the monomers, assigning charge groups and performing energy minimization.

Each model comprised a single atactic homopolymer chain (20 repeat units), constructed using conditional torsional statistics and containing approximately $2.7\text{--}5.3 \times 10^3$ atoms depending on side-chain length. For each MPIL variation (ethyl, butyl, octyl), we generated $n = 3$ independent replicas by reseeding torsional states and packing (nine simulation boxes in total). System compositions are reported in Table S2.

All chains were energy-minimized to remove unrealistic internal strain (e.g., torsional) and to obtain relaxed starting conformations. The minimized chain was embedded in a periodic cubic cell with explicit water and ions. Amorphous models were constructed at 298 K by solvating the polymer with 151–4882 water molecules (according to the

target hydration level) and adding the appropriate ions to maintain electroneutrality (BF_4^- counterions to the phosphonium sites; Na^+ and Cl^-). Cubic periodic boundary conditions (PBC) were applied to all systems to emulate the bulk hydrated membrane.

Water molecules were added based on experimentally determined water uptake (WU) values [37]: 1) MPIL_octyl: 9 wt%, 2) MPIL_butyl: 27 wt%, 3) MPIL_ethyl: 81 wt%. Water uptake was calculated as the percentage increase in membrane mass upon hydration, defined as: $\text{WU} = \frac{W(\text{wet}) - W(\text{dry})}{W(\text{dry})} \times 100\%$, WU is calculated from the “wet” weight, $W(\text{wet})$, at each equilibrium and the “dry” weight, $W(\text{dry})$, at the end of the initial drying step. The detailed composition of the PBC boxes is provided in Table S2, and three independent models were constructed for each MPIL polymer to ensure reproducibility and statistical accuracy. Each system was constructed in a box of $\sim 38\text{--}56$ Å per side, leading to a total of 5000–17,000 atoms.

Table 1

Monomer structures and 3D models of MPIL_octyl, MPIL_butyl, and MPIL_ethyl hydrated systems. Atom types are distinguished by color: C (gray), H (white), O (red), B (pink), F (light blue), P (purple), Na (dark purple), and Cl (green).

Name	Chemical structure	3-D Models
MPIL_Octyl (-C ₈ H ₁₇)		
MPIL_Butyl (-C ₄ H ₉)		
MPIL_Ethyl (-C ₂ H ₅)		



2.2. Computational details of MD simulation

Chain segments were distributed using a combination of the Theodorou-Suter approach and Meirovitch's scanning method to ensure a realistic distribution of torsional angles. PBC boxes were generated following a ramp density protocol, starting at 1.05 g/cm³. The models then underwent equilibration procedures, followed by geometry optimization using the Smart algorithm (Energy: 0.001 kcal/mol, Force: 0.5 kcal/mol/Å, Max iterations: 5000). Subsequently, molecular dynamics simulations were carried out in the isothermal–isobaric ensemble (1 atm, 293 K) for 3 ns with a 1.0 fs time step. The COMPASS III force field was used, with long-range electrostatic interactions calculated using a particle-particle-particle mesh algorithm (accuracy: 0.001 kcal/mol) and atom-based van der Waals interactions with a 12.5 Å cutoff.

Temperature and pressure were controlled via the Berendsen thermostat, with a decay constant of 0.1 ps [58]. Equilibration was assessed by monitoring density fluctuations over time, confirming that the system reached a stable density (energy) at the end of the 3 ns simulation. Since we were unable to obtain experimental density values for the humidified models due to limitations in the buoyancy-based density measurement method, some uncertainty remains regarding the absolute accuracy of the equilibrated structures.

The evolution of the total energy during the 20 ns production run is shown in Fig. S2, confirming that the systems remained stable throughout the simulation. To analyze the behavior of the models, a subsequent 20 ns at 293 K MD simulation was conducted. Detailed analyses of the behavior of the polymer, ions, and water molecules were performed at 20 ns to assess their dynamic interactions and stability within the system.

In simulations, single-chain amorphous cells with periodic boundaries were used, so membrane thickness is not defined at the atomistic scale. Experimentally, membranes were ~ 100 μm thick and crosslinked via diacrylate monomer, as detailed in Ref. [37]. In the computational models, no explicit crosslink bonds were introduced, to isolate the role of alkyl chain length on hydration and ion transport.

2.3. Structural characterization: slice-based visualization

To analyze the molecular-level organization of MPIL membranes, we examined water distribution, polymer chain arrangement, and ion positioning at the end of the simulation runs. For qualitative visualization of morphology, the simulation boxes were displayed as slices of ~4 Å thickness. As these are amorphous bulk systems with 3D periodic boundary conditions, no preferential layering along the z-axis is implied; the slices are purely a graphical tool to highlight water clustering and connectivity.

2.4. Diffusion coefficients for ions

The ionic transport is characterized via MSD-derived diffusivities. The diffusion coefficients of Cl[−] ions were calculated based on mean square displacement (MSD) data obtained from the MD simulations. The MSD and diffusion coefficients were determined using the following equations:

$$\text{MSD} = \frac{1}{N} \left\langle \sum_{i=1}^N |r_i(t) - r_i(0)|^2 \right\rangle = \Delta |r(t)|^2 \quad (1)$$

$$D = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d}{dt} \left\langle |r_i(t) - r_i(0)|^2 \right\rangle \quad (2)$$

where $r_i(t)$ and $r_i(0)$ are the position vectors of ion i at times t and 0, respectively, during the MD simulation. The diffusion coefficient (D) was calculated based on the Einstein equation [58]. The calculated diffusion coefficients were compared with experimental Cl[−] conductivity data to evaluate the membrane's performance.

More rigorous conductivity estimators (e.g., Einstein–Helfand/Green–Kubo) typically require substantially longer trajectories to converge charge-current correlations, while Nernst–Einstein values provide an upper bound as they neglect ion–ion correlations.

2.5. Interaction between functional groups and ions in membrane

The interactions between phosphonium functional groups and ions were analyzed using the radial distribution function (RDF) and coordination number, $n(r)$, calculations. The RDF represents the particle density distribution at a specific distance, calculated using the following equation:

$$g(r) = \frac{dn_r}{4\pi r^2 dr \times \rho} \quad (3)$$

where $g(r)$ is the RDF, dn_r represents the average number of particles contained within a spherical shell centered on the reference particle. r is the radial distance, and ρ is the average particle density. The $n(r)$ was determined by integrating $g(r)$ up to a cutoff radius r :

$$n(r) = 4\pi\rho \int_0^r g(r')r'^2 dr' \quad (4)$$

Coordination numbers were integrated up to the first minimum of $g(r)$. These analyses provided insights into the local environments of ions and their interactions with polymer functional groups.

2.6. Computational details of pore analysis

The dry (non-hydrated) membrane structures, extracted from the last 5 ns of the MD trajectories, were analyzed with Zeo++ software package [59]. The accessible pore volume was computed using a probe radius of 1.40 Å, corresponding to the kinetic radius of a water molecule, which is a standard convention for water accessibility analyses in porous and polymeric systems. For each MPIL variant (ethyl, butyl, octyl), three independent replicas were converted to CIF format and processed with the -vol routine of Zeo++. The following descriptors were obtained: 1) Accessible volume (AV, Å³): total probe-occupiable volume within the simulation cell. 2) Accessible volume fraction (AV/Vcell): ratio of accessible volume to the unit cell volume. 3) Number of channels: continuously percolated pathways detected by the probe. All results were averaged over three replicas per system, and uncertainties are reported as standard deviations.

2.7. Ion association free energy from RDF

The thermodynamics of ion pairing between the quaternized phosphonium center (P⁺) and the BF₄[−] counterion were obtained from the radial distribution function $g(r)$ [60–62]. The contact-ion-pair cutoff r_c was defined as the first minimum following the contact-ion-pair peak. The configurational equilibrium constant was evaluated as

$$K_{\text{vol}} = 4\pi \int_0^{r_c} r^2 g(r) dr \quad (5)$$

which has units of volume. Conversion to the 1 M standard state used $V^0 = 1.660 \text{ nm}^3$ (corresponding to the molecular volume at 1 M concentration), yielding $K^0 = K_{\text{vol}}/V^0$. The standard free energy of association was then $\Delta G^0 = -RT \ln K^0$, with $R = 8.314 \times 10^{-3} \text{ kJ mol}^{-1} \text{ K}^{-1}$ and $T = 298 \text{ K}$. Uncertainties were estimated from replicate simulations (three per family) and from varying r_c by $\pm 0.02 \text{ nm}$.

2.8. Experimental MPIL membrane preparation and characterization

The membranes were prepared from homogenous mixture of polymerizable ionic liquids as described in a previous work [37]. Shortly,

each of the three phosphonium-based polymerizable ionic liquids containing a styrenic functionality (Fig. S1) is mixed with an acrylate monomer, a di-acrylate crosslinker and a photo-initiator to produce by casting and successive photopolymerization dense membranes.

The membrane conductivity was measured by electrochemical impedance spectroscopy (EIS) in NaCl 0.5 M at 298 K by a potentiostat/galvanostat combined with a frequency response analyzer (Metrohm Autolab PGSTAT302N) in the frequency range 1000–0.01 Hz, with a signal amplitude of 10 mV [63].

3. Results and discussion

3.1. Ionic conductivity measurements and diffusion coefficients calculations

The ionic conductivity of the MPIL membranes (Fig. 1 and Table 2) clearly increases with decreasing alkyl side-chain length, following the order MPIL_ethyl > MPIL_butyl > MPIL_octyl. For MPIL_ethyl and MPIL_butyl, the error bars are clearly visible, reflecting larger fluctuations between replicas, whereas for MPIL_octyl the standard deviation is very small, and the error bars are almost hidden by the marker. This high reproducibility highlights the statistical robustness of the observed chain-length dependence. The conductivity trend is consistent with enhanced hydration and the formation of more interconnected aqueous pathways for shorter side chains. It also aligns with expected microstructural effects, such as hydrophilic domain size, tortuosity, and interconnectivity, that govern transport. Notably, the ionic conductivity of MPIL_ethyl exceeded that of the benchmark commercial AEM Selemion AMVN (AGC, USA), underscoring the potential of phosphonium-based PILs as next-generation energy membranes. A detailed experimental study of structure–property relationships in phosphonium-based AEMs, including these materials, is underway [26].

To quantify ion mobility in hydrated MPIL membranes, we calculated Cl^- diffusion coefficients from MSD. The 5 K difference between simulation (293 K) and experiment (298 K) does not alter the diffusivity hierarchy: a 293 → 298 K Arrhenius sensitivity (SI, Table S3) yields a modest scaling factor of ≈ 1.07 – 1.15 for plausible activation energies, with no change in ordering (ethyl > butyl > octyl). Because absolute

Table 2

Ionic conductivity (σ , $\text{mS}\cdot\text{cm}^{-1}$) of MPIL membranes compared with a commercial anion-exchange membrane (Selemion AMVN). Reported values are mean $\pm$ standard deviation from three independent samples; the benchmark value is provided as manufacturer specification.

Membrane	Conductivity σ (mS/cm)
MPIL_octyl	$(4.85 \pm 0.07) \times 10^{-2}$
MPIL_butyl	$(1.22 \pm 0.03) \times 10^0$
MPIL_ethyl	$(8.84 \pm 0.01) \times 10^0$
Selemion AMVN	5.00 (benchmark, no error)

conductivities inferred via Nernst–Einstein can be biased upward by neglected ion–ion correlations, we emphasize comparative trends rather than absolute values. More rigorous conductivity estimators (e.g., Einstein–Helfand [64] capture charge–current correlations but typically require substantially longer trajectories to converge. The consistency between MSD-derived trends and EIS data supports the robustness of the structure–property relationships reported. Fig. S3 shows MSDs over the final 10 ns of production MD; Table 3 summarizes the resulting D_{Cl^-} values.

The increase in diffusion coefficient agrees with the experimental conductivity trends in Fig. 1. The results confirm that shorter alkyl chains promote higher WU, which in turn facilitates greater ion mobility.

The diffusion coefficient increases significantly from MPIL_octyl to MPIL_butyl, with a 30-fold increase, demonstrating the critical impact of hydration on the ion transport. MPIL_ethyl exhibits the highest

Table 3

Cl^- diffusion coefficients (D , $\text{\AA}^2\cdot\text{ps}^{-1}$) calculated from MSD analysis of hydrated MPIL membranes. Values are averaged over three independent replicas; uncertainties represent standard deviations. diffusion coefficients.

Membrane	D_{Cl^-} ($\text{\AA}^2\cdot\text{ps}^{-1}$)
MPIL_octyl	$(5.70 \pm 4.50) \times 10^{-4}$
MPIL_butyl	$(1.75 \pm 0.09) \times 10^{-2}$
MPIL_ethyl	$(5.50 \pm 1.20) \times 10^{-1}$

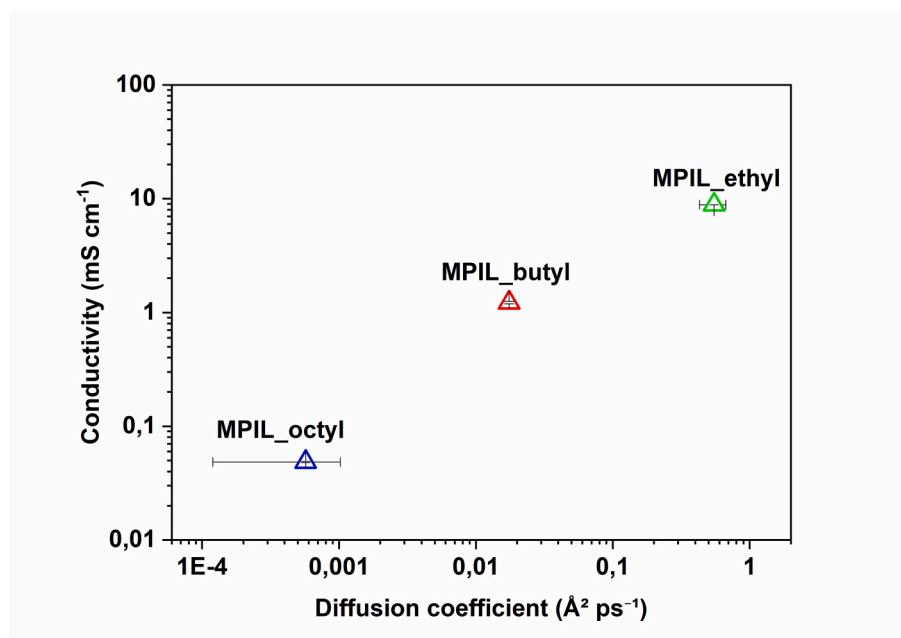


Fig. 1. Correlation between simulated and experimental transport properties. x-axis: MD-derived Cl^- diffusion coefficients ($\text{\AA}^2\cdot\text{ps}^{-1}$, mean $\pm$ standard deviation, three replicas). y-axis: EIS conductivity ($\text{mS}\cdot\text{cm}^{-1}$, 0.5 M NaCl, mean $\pm$ standard deviation, three independent membranes). Error bars indicate MD statistical uncertainty (horizontal) and experimental reproducibility (vertical). Data points: MPIL_ethyl, MPIL_butyl, MPIL_octyl.

diffusion, supported by its extensive hydration networks. This analysis highlights a membrane design trade-off: MPIL_{octyl} limits ion diffusion through localized hydrophilic domains, indicating a stronger tendency for ion retention and reduced overall mobility. MPIL_{ethyl} optimizes ion conductivity, but may increase counterion permeability, potentially affecting the ion retention tendency. MPIL_{butyl} represents a balanced case, offering both reasonable conductivity and moderate mobility control, making it suitable for applications requiring a compromise between the two properties.

3.2. Morphological analysis of MPIL polymers via MD simulations

MD simulations reveal how polymer architecture affects hydration and ion transport (Fig. 2).

For 3D models, water molecules are shown in blue, sodium ions (Na^+) in purple, chloride ions (Cl^-) in bright green, BF_4^- ions in dark green, and the polymer in gray.

MPIL_{octyl}, with the longest chains, exhibited the lowest water uptake, forming small and confined hydrophilic regions. MPIL_{ethyl}, with the shortest chains, showed extensive hydration, forming highly interconnected hydrophilic domains that enhance ion transport. MPIL_{butyl} displayed an intermediate behavior, balancing water uptake and structural stability.

Fig. 3 represents slices of ~ 4 Å thickness, highlighting how water molecules (blue regions) are spatially organized and interconnected within the polymer matrix and illustrates the morphological differences among the three MPILs, emphasizing how chain length impacts water channel formation. The slices illustrate morphological features (water clusters, polymer-ion contacts); they do not indicate stratification along z, which is absent in isotropic bulk PBC models.

As illustrated in Fig. 3, decreasing the alkyl side chain length increases water interconnectivity. In MPIL_{octyl}, the low water uptake results in dispersed water molecules forming small, isolated clusters. As water uptake increases, water domains progressively interconnect, becoming fully percolated in MPIL_{ethyl}.

The extended alkyl chains of MPIL_{octyl} create steric hindrance, limiting the formation of hydrophilic domains and restricting ion mobility. In contrast, the MPIL_{ethyl} shorter chains facilitate the formation of extensive, interconnected hydrophilic regions, enabling superior ion transport. MPIL_{butyl} represents an intermediate case, balancing water uptake with structural stability.

These morphological differences confirm that water uptake strongly correlates with ion transport efficiency [65–71]. These findings highlight the importance of optimizing polymer architecture to achieve desired performance metrics for specific applications.

Zeo++ analysis with a 1.4 Å probe radius (Table 4) further quantified channel connectivity, confirming that MPIL_{ethyl} exhibits a fully percolated aqueous domain (Accessible Volume $\approx 1.26 \times 10^5 \text{ Å}^3$, $\sim 72\%$ of the box volume), consistent across replicas. MPIL_{butyl} showed partial percolation with a much smaller accessible fraction ($\sim 8\%$), while MPIL_{octyl} contained only isolated voids with negligible accessible volume and no percolated channels. These quantitative results corroborate the conductivity trends observed experimentally (ethyl > butyl > octyl). The probe radius was selected following the standard convention for water accessibility analyses in porous materials and membranes [59,72]. The accessible volume fractions obtained from Zeo++ (0 % for MPIL_{octyl}, $\sim 8\%$ for MPIL_{butyl}, and $\sim 72\%$ for MPIL_{ethyl}) are fully consistent with the experimental water uptake values ($\approx 9, 27$, and $81 \text{ wt } \%$, respectively). Although water uptake is expressed as mass increase and Zeo++ descriptors quantify the probe-accessible void fraction, both

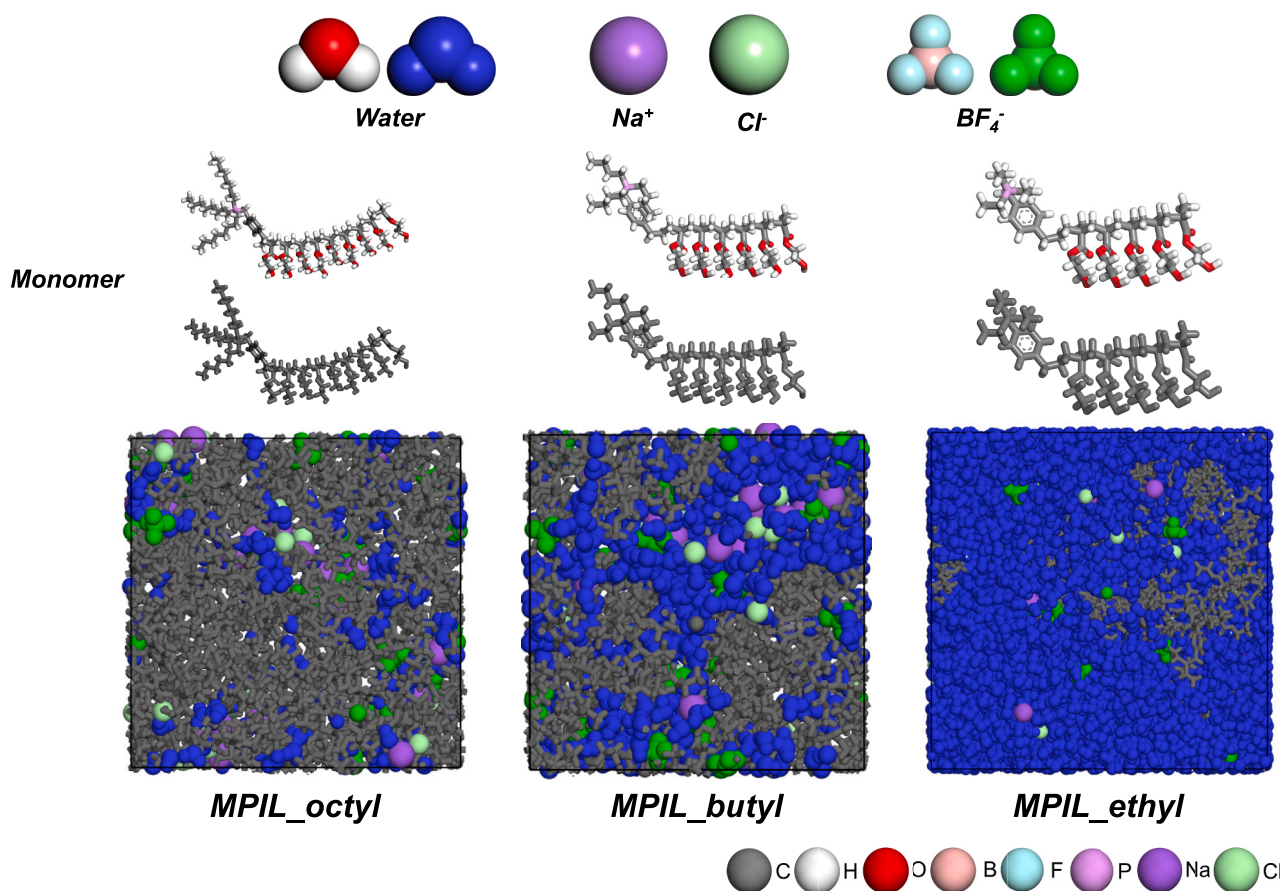


Fig. 2. Representative snapshots of hydrated MPIL polymers. Atom types for monomers are distinguished by color: C (gray), H (white), O (red), B (pink), F (light blue), P (purple), Na (dark purple), and Cl (green).

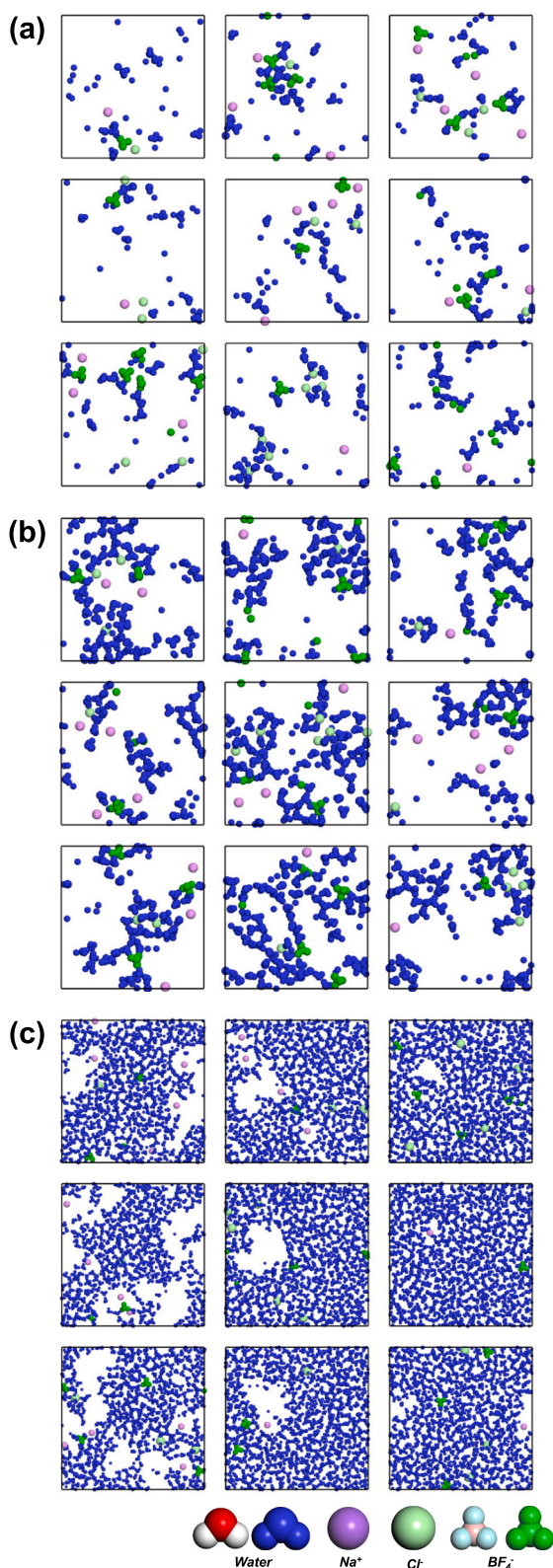


Fig. 3. Qualitative slice visualization (~ 4 Å thickness) of hydrated MPIL models, illustrating water distribution and channel formation as a function of alkyl side chain length: (a) MPIL_{octyl}, (b) MPIL_{butyl}, and (c) MPIL_{ethyl}. As the alkyl chain length decreases, water domains become more interconnected, forming continuous channels in MPIL_{ethyl}. Water molecules are shown in blue, sodium ions (Na^+) in purple, chloride ions (Cl^-) in bright green, BF_4^- ions in dark green.

Table 4

Zeo++ pore connectivity parameters (probe radius = 1.4 Å) for hydrated MPIL membranes. Values are averaged over three independent replicas; uncertainties represent standard deviations.

Membrane	Accessible volume (Å ³)	Accessible fraction	Channels
MPIL _{octyl}	0 ± 0	0.000 ± 0.000	0
MPIL _{butyl}	(3912 ± 90)	0.080 ± 0.002	1
MPIL _{ethyl}	(125,565 ± 510)	0.725 ± 0.002	1

measures confirm the same hierarchy of hydration and connectivity (ethyl > butyl > octyl).

3.3. Interaction between phosphonium groups and ions/water

The RDF analysis over the last 10 ns quantifies ion and water interactions with phosphonium groups (Table 5).

Despite MPIL_{octyl}'s high hydrophobicity, ions and water molecules remain confined near phosphonium groups (shorter RDF peaks), forming localized hydration shells. This restricts ion diffusion, correlating with lower conductivity. The strong polymer-ion interactions (short RDF distances for Cl^- and BF_4^-) indicate that ions preferentially remain near the charged groups rather than distributing freely throughout the polymer matrix. The hydrophobicity of the long alkyl chains (C8) limits water penetration, preventing the formation of extended hydration networks. Water molecules also remain localized near the phosphonium groups, forming small hydration shells rather than continuous water pathways. As a result, this strong confinement reduces ion mobility, leading to lower conductivity but higher ion retention tendency making MPIL_{octyl} suitable for applications where controlled ion retention is needed. In contrast, MPIL_{ethyl} exhibits longer RDF distances, confirming broader ion dispersion and more extensive hydration networks, leading to higher ion mobility. With shorter alkyl chains (C2), MPIL_{ethyl} is highly hydrophilic, promoting extensive hydration and water uptake. The weaker polymer-ion interactions (longer RDF distances) allow Cl^- and BF_4^- to migrate further from the phosphonium groups, spreading more uniformly throughout the hydrated structure. The larger RDF peak for water suggests that water molecules are less localized, forming interconnected hydration networks instead of isolated hydration shells. In conclusion this enhanced ion mobility and water channel formation explain the higher ionic conductivity observed in MPIL_{ethyl}. MPIL_{butyl} falls between these extremes, maintaining controlled hydration while supporting moderate ion transport.

Figs. 4 and S4 provide a comprehensive analysis of the interaction between phosphonium groups and ions/water across different MPIL polymers. Fig. 4(a) displays the RDF curves for Cl^- , BF_4^- , and water molecules relative to the phosphonium headgroups in MPIL_{octyl}, MPIL_{butyl}, and MPIL_{ethyl}. The RDF peaks indicate the preferred spatial arrangement of ions and water molecules within the polymer matrix.

Fig. 4(b) quantifies the average number of ions and water molecules surrounding the phosphonium groups within a given cutoff distance. MPIL_{octyl} exhibits the highest $n(r)$, indicating dense ion packing and

Table 5

RDF first peaks between phosphonium and ions/water.

Polymer	Ion/water	RDF first peak distance (Å)
MPIL _{octyl}	Cl^-	4.67 ± 0.15
	BF_4^-	4.53 ± 0.01
	Water	3.71 ± 0.08
MPIL _{butyl}	Cl^-	5.28 ± 0.14
	BF_4^-	5.06 ± 0.24
	Water	5.00 ± 0.02
MPIL _{ethyl}	Cl^-	5.80 ± 0.15
	BF_4^-	5.41 ± 0.19
	Water	5.09 ± 0.02

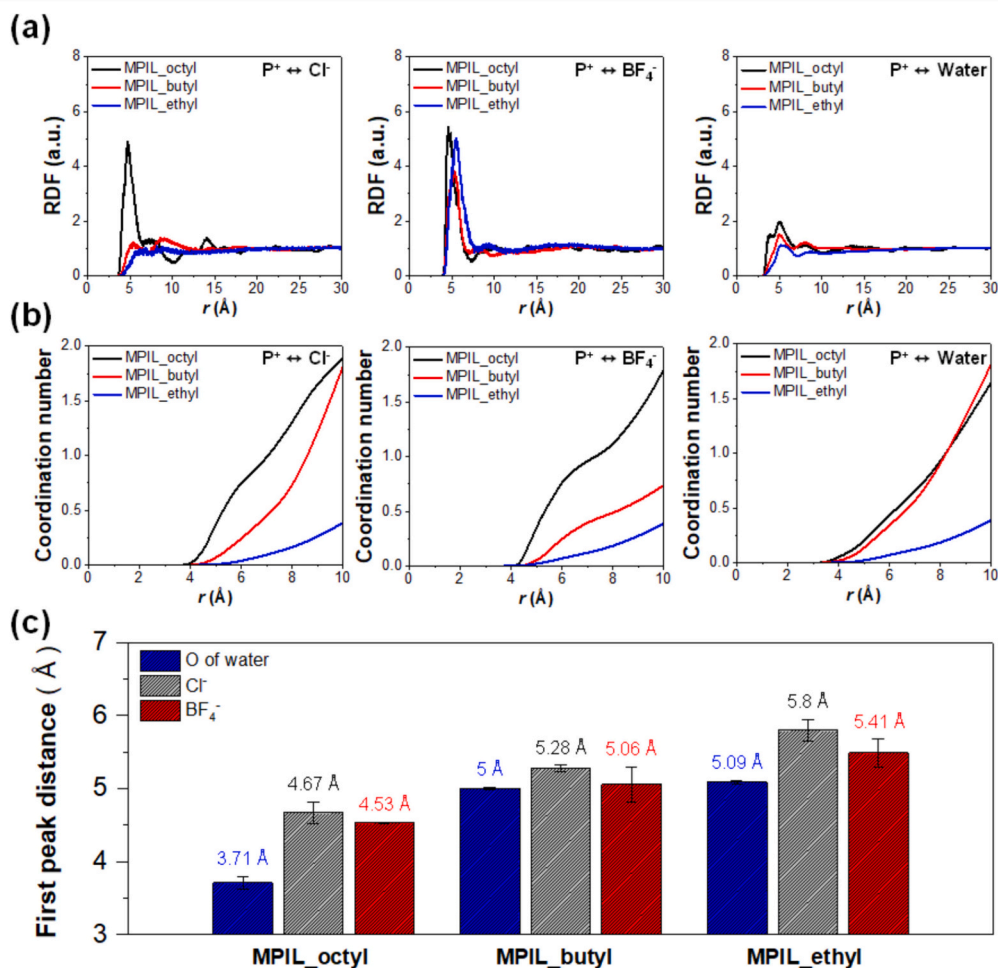


Fig. 4. RDF between phosphonium groups and ions/water relating MPIL_octyl, MPIL_butyl and MPIL_ethyl: (a) radial distribution function, (b) coordination number, and (c) distance of phosphonium cations and ions/water at 20 ns.

strong electrostatic interactions near the phosphonium sites and oppositely MPIL_ethyl has the lowest $n(r)$, confirming that ions and water molecules are more dispersed and less tightly bound to the phosphonium groups. Finally, Fig. 4(c) shows the evolution of ion and water distances over time.

Integration of the $P^+-BF_4^-$ RDF up to the first minimum yields standard association free energies of -2.20 , -1.22 , and -0.86 $\text{kJ}\cdot\text{mol}^{-1}$ for MPIL_ethyl, MPIL_octyl, and MPIL_butyl, respectively (averaged over three replicas). These results (Table S4) show that shorter chains (MPIL_ethyl) leave the cationic center more exposed, favoring stronger ion pairing, whereas intermediate and longer chains (MPIL_butyl, MPIL_octyl) partially shield the charged site, thereby reducing the stability of the ion pair. The trend is not strictly monotonic, as MPIL_octyl introduces nanophase segregation that can promote slightly more stable ion contacts compared to MPIL_butyl. Overall, although the absolute differences are modest (~ 1.3 $\text{kJ}\cdot\text{mol}^{-1}$ across the series), the effect is systematic and consistent with the expected screening imposed by longer alkyl substituents on the quaternized phosphonium group.

3.4. Side chain length and hydration dynamics in MPIL polymers

The influence of phosphonium side chain length on polymer hydration and ion transport was quantified, as shown in Table 6.

The data reveals that side chain length systematically decreases from MPIL_octyl to MPIL_ethyl. In MPIL_octyl, the long chains limit water

Table 6

Phosphonium chain length in MPIL Polymers.

Polymer	Side chain	Average length (Å)
MPIL_octyl	$-C_8H_{17}$	8.88 ± 0.04
MPIL_butyl	$-C_4H_9$	5.14 ± 0.02
MPIL_ethyl	$-C_2H_5$	2.82 ± 0.004

accessibility, reducing ion mobility but enhancing the tendency for ion retention. Our observation that longer alkyl side chains reduce hydration connectivity by sterically shielding polar sites is consistent with prior evidence that self-association or steric crowding diminishes the availability of H-bond donors/acceptors, thereby lowering water uptake despite nominal polarity [73]. Conversely, the MPIL_ethyl shorter chain facilitates continuous hydration networks, improving conductivity. MPIL_butyl, with an intermediate chain length, balances hydration and ion mobility, making it versatile for applications requiring a trade-off between mobility and retention control.

3.5. Interaction between polymer backbone and water

To evaluate the role of backbone functional groups in hydration and ion transport, water interactions with ester and hydroxyl groups were analyzed. Fig. S5 indicates the results of radial distribution function between the backbone functional groups and water molecules. Table 7

Table 7

Water interaction with polymer backbone functional groups.

Polymer	Backbone functional group	Average distance with water molecules (Å)
MPIL _{octyl}	–C ₈ H ₁₇	
	Carbonyl	1.78 ± 0.01
	Alkoxy linkage	3.98 ± 0.02
	–OH group (O)	1.64 ± 0.01
MPIL _{butyl}	–C ₄ H ₉	
	Carbonyl	1.74 ± 0.01
	Alkoxy linkage	4.00 ± 0.02
	–OH group (O)	1.62 ± 0.01
MPIL _{ethyl}	–C ₂ H ₅	
	Carbonyl	1.72 ± 0.01
	Alkoxy linkage	5.85 ± 0.04
	–OH group (O)	1.62 ± 0.01
	–OH group (H)	1.78 ± 0.01

summarizes RDF first peaks distances based on Fig. S5.

The hydroxyl groups exhibited the strongest interactions with water due to their high polarity and ability to form hydrogen bonds. These interactions play a pivotal role in maintaining hydration and promoting water retention within the polymer matrix. In MPIL_{ethyl}, a reduced steric hindrance allowed for stronger interactions between water molecules and carbonyl groups, enhancing hydration and water channel formation. The alkoxy linkage groups, in contrast, demonstrated weaker interactions with water, with interaction distances increasing in MPIL_{L_ethyl} due to the polymer open structure. These findings emphasize the significance of polymer architecture in modulating hydration dynamics and optimizing ion transport pathways.

3.6. Summary of structural insights

The structural properties of MPIL polymers are strongly influenced by side chain length, which governs hydration dynamics, ion transport, and membrane performance. Fig. 5 visually represents these structural differences for MPIL_{octyl} (a), MPIL_{butyl} (b), and MPIL_{ethyl} (c), highlighting how side chain length impacts water distribution, ion mobility, and hydrogen bonding.

In MPIL_{octyl} (Fig. 5(a)), the long *n*-alkyl chains create steric hindrance, forming hydrophobic barriers that restrict water penetration and confine hydration to localized domains near the phosphonium groups (purple spheres). This limited hydration results in strong polymer-ion interactions, keeping counterions such as Cl[−] (green spheres) in proximity to the charged headgroups. The small hydration shells (blue dashed circles) indicate that water molecules are less available to form extended hydrogen bonding networks, leading to lower ion mobility but higher ion retention tendency.

MPIL_{butyl} (Fig. 5(b)) represents an intermediate case, where hydration is more extensive than in MPIL_{octyl} but not as widespread as in MPIL_{ethyl}. The shorter *n*-alkyl chains reduce steric hindrance, allowing

water to access functional groups more freely. This results in larger hydration regions and a more open polymer structure. Partial water channel formation improves ion mobility while still maintaining some degree of retention tendency, making MPIL_{butyl} a balanced system between ion conduction and structural integrity.

In MPIL_{ethyl} (Fig. 5(c)), the short *n*-alkyl chains minimize steric hindrance, enabling extensive hydration and continuous aqueous networks. Water molecules (red and white spheres) are more uniformly distributed, forming well-connected hydrogen bonding networks (blue dashed lines). This highly hydrated environment weakens polymer-ion interactions, allowing counterions to disperse throughout the polymer matrix rather than being confined near the phosphonium groups. As a result, MPIL_{ethyl} exhibits the highest ionic conductivity, making it ideal for applications requiring efficient ion transport, such as fuel cells and water electrolysis.

Overall, side chain length critically influences water accessibility, ion mobility, and membrane conductivity.

MPIL_{octyl} restricts ion transport due to its hydrophobicity, MPIL_{L_ethyl} maximizes conductivity through extensive hydration, and MPIL_{butyl} provides a compromise between mobility and retention control. This structural variability allows for the design of MPIL-based membranes tailored to specific applications, balancing conductivity and ion retention as needed.

4. Conclusion

This work combines experimental measurements and atomistic molecular dynamics simulations to elucidate how the *n*-alkyl chain length in phosphonium-based polymerized ionic liquids (MPIs) governs the performance of anion exchange membranes (AEMs). The results establish a clear structure–property relationship linking chain length to hydration dynamics, ion mobility, and microstructural organization.

Shorter side chains (MPIL_{ethyl}) promote extensive water uptake and form highly interconnected hydrophilic domains, enabling the highest ionic conductivity and fastest ion transport. Notably, MPIL_{ethyl} achieves a conductivity of 8.84 mS·cm^{−1}, surpassing that of the benchmark commercial Selemion AMVN membrane (5.0 mS·cm^{−1} under identical conditions), thereby demonstrating practical competitiveness of phosphonium-based PILs. Conversely, longer chains (MPIL_{L_ethyl}) limit water accessibility and confine ion transport to localized hydration regions, enhancing ion retention at the expense of conductivity. The intermediate chain length (MPIL_{butyl}) achieves a tunable compromise between conductivity and retention, offering flexibility for applications that require both moderate ion mobility and controlled counterion retention. While no direct experimental measurements of selectivity were performed in this study, the MD-derived morphological trends indicate relative differences in ion retention and mobility control that may be validated by future targeted experiments (e.g., migration number analysis).

Molecular-level analyses consistently support these performance trends. Connectivity analysis revealed that only MPIL_{ethyl} exhibits a

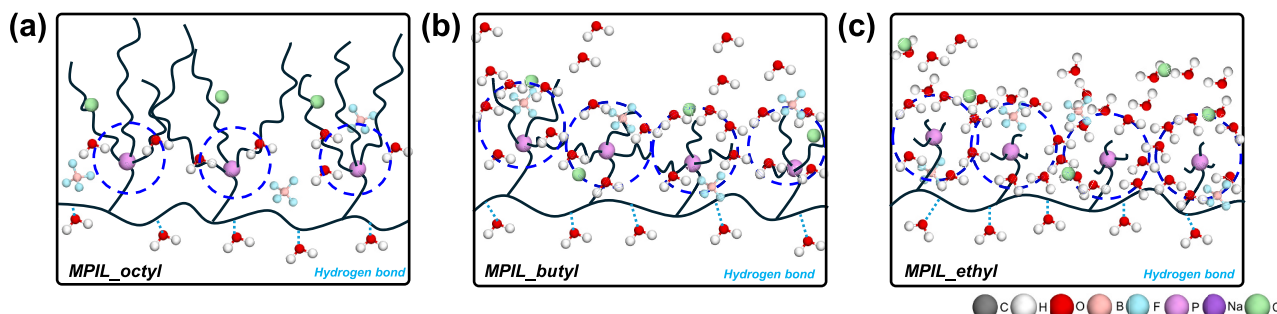


Fig. 5. Illustrations of the behavior of water and ions in the MPIL polymer according to chain length. (a) MPIL_{octyl}, (b) MPIL_{butyl}, and (c) MPIL_{ethyl}.

fully percolated aqueous network, while MPIL-butyl shows partial connectivity and MPIL-octyl lacks continuous pathways. Ion association free energies derived from RDF integration confirmed stronger pairing for ethyl due to its exposed cationic centers, with steric shielding from longer chains reducing ion association stability. Backbone–water RDFs further highlighted the critical role of hydroxyl and carbonyl groups in sustaining hydration and enabling water channel formation. Together with Arrhenius sensitivity analysis, these results reinforce the robustness of the computational–experimental correlation.

The insights provide practical design guidelines: by adjusting alkyl chain length, it is possible to deliberately target high-conductivity, high mobility control, or balanced-performance membranes, depending on the operational requirements.

Future work will extend this design framework to include additional functional group modifications, varied hydration levels, and different operating environments, further enabling the rational engineering of MPIL-based AEMs for next-generation electrochemical energy conversion and separation technologies.

CRedit authorship contribution statement

Jun Ho Park: Formal analysis, Data curation, Investigation, Visualization, Writing-original draft,

Enrica Fontananova: Investigation, Writing – review.

Francesco Galiano: Data Curation.

Alberto Figoli: Data Curation.

Raffaella Mancuso: Data Curation.

Bartolo Gabriele: Writing – review.

Tae Kyung Lee: Conceptualization, Software, Writing – review & editing.

Chi Hoon Park: Funding acquisition.

Elena Tocci: Supervision, Conceptualization, Writing – review & editing.

Sang Yong Nam: Supervision, Funding acquisition, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by:

A grant from the Korea Institute for Advancement of Technology (KIAT), funded by the Korean Government (MOTIE) (P0017310, Human Resource Development Program for Industrial Innovation (global)).

A grant under the framework of the international cooperation program, managed by the National Research Foundation of Korea (NRF) (NRF-2023K2A9A1A06094902, FY2023).

A grant from the Basic Science Research Program through the National Research Foundation of Korea (NRF), funded by the Ministry of Education (2020R1A6A03038697).

The Consiglio Nazionale delle Ricerche of Italy is gratefully acknowledged for its financial support of the project ACCORDO CNR-NRF 2024–2025 on “Novel anion exchange ionomers and membranes for alkaline fuel cells and water electrolyzers: a computational and experimental development.”

Further support was provided by the European Union – Next Generation EU, within the framework of the project PNRR - AdP IDROGENO (PRR.AP015.017.002); it has also received funding from the Italian Ministry of Environment and Energetic Security (MASE) in the framework of the projects System Research - Frontier material for energy use (PTR_25_27_CNR_1_4).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.desal.2025.119434>.

Data availability

Data will be made available on request.

References

- [1] S. Chu, A. Majumdar, Opportunities and challenges for a sustainable energy future, *Nature* 488 (2012) 294–303, <https://doi.org/10.1038/nature11475>.
- [2] R.W. Baker, *Membrane Technology and Applications*, John Wiley & Sons, Ltd, 2012.
- [3] A.I. Osman, Z. Chen, A.M. Elgaray, M. Farghali, I.M.A. Mohamed, A.K. Priya, H. B. Hawash, P.S. Yap, Membrane technology for energy saving: principles, techniques, applications, challenges, and prospects, *Adv. Energy Sustain Res.* 5 (2024), <https://doi.org/10.1002/aesr.202400011>.
- [4] E. Drioli, E. Tocci, F. Macedonio, Membrane science and membrane engineering: a successful story, *Gen. Chem.* 5 (2019) 180027–180027, <https://doi.org/10.21127/yaoyigc20180027>.
- [5] S.P. Bera, M. Godhaniya, C. Kothari, Emerging and advanced membrane technology for wastewater treatment: a review, *J. Basic Microbiol.* 62 (2022) 245–259, <https://doi.org/10.1002/jobm.202100259>.
- [6] T. Luo, S. Abdu, M. Wessling, Selectivity of ion exchange membranes: a review, *J. Membr. Sci.* 555 (2018) 429–454, <https://doi.org/10.1016/j.memsci.2018.03.051>.
- [7] J. Ran, L. Wu, Y. He, Z. Yang, Y. Wang, C. Jiang, L. Ge, E. Bakangura, T. Xu, Ion exchange membranes: new developments and applications, *J. Membr. Sci.* 522 (2017) 267–291, <https://doi.org/10.1016/j.memsci.2016.09.033>.
- [8] S. Jiang, H. Sun, H. Wang, B.P. Ladewig, Z. Yao, A comprehensive review on the synthesis and applications of ion exchange membranes, *Chemosphere* 282 (2021), <https://doi.org/10.1016/j.chemosphere.2021.130817>.
- [9] S. Depuydt, B. Van der Bruggen, Green synthesis of cation exchange membranes: a review, *Membranes* 14 (2024), <https://doi.org/10.3390/membranes14010023>.
- [10] A.K. Thakur, M. Malmali, Advances in polymeric cation exchange membranes for electrodialysis: an overview, *J. Environ. Chem. Eng.* 10 (2022) 108295, <https://doi.org/10.1016/j.jece.2022.108295>.
- [11] E.Y. Safronova, D.V. Golubenko, N.V. Shevlyakova, M.G. D'yakova, V.A. Tverskoi, L. Dammak, D. Grande, A.B. Yaroslavtsev, New cation-exchange membranes based on cross-linked sulfonated polystyrene and polyethylene for power generation systems, *J. Membr. Sci.* 515 (2016) 196–203, <https://doi.org/10.1016/j.memsci.2016.05.006>.
- [12] P.X. Yang, J. Wang, H.L. Liu, Z.Y. Guo, Z.H. Huang, P.P. Zhang, Z.Y. Ji, High-performance monovalent selective cation exchange membranes with ionically cross-linkable side chains: effect of the acidic groups, *ACS Appl. Mater. Interfaces* 16 (2024) 35576–35587, <https://doi.org/10.1021/acsami.4c07085>.
- [13] G.H.A. Wijaya, K.S. Im, S.Y. Nam, Advancements in commercial anion exchange membranes: a review of membrane properties in water electrolysis applications, *Desalin. Water Treat.* 320 (2024) 100605, <https://doi.org/10.1016/j.dwt.2024.100605>.
- [14] V. Vijayakumar, S.Y. Nam, Recent advancements in applications of alkaline anion exchange membranes for polymer electrolyte fuel cells, *J. Ind. Eng. Chem.* 70 (2019) 70–86, <https://doi.org/10.1016/j.jiec.2018.10.026>.
- [15] N. Du, C. Roy, R. Peach, M. Turnbull, S. Thiele, C. Bock, Anion-exchange membrane water electrolyzers, *Chem. Rev.* 122 (2022) 11830–11895, <https://doi.org/10.1021/acs.chemrev.1c00854>.
- [16] K.F.L. Hagestijn, S. Jiang, B.P. Ladewig, A review of the synthesis and characterization of anion exchange membranes, *J. Mater. Sci.* 53 (2018) 11131–11150, <https://doi.org/10.1007/s10853-018-2409-y>.
- [17] C. Zheng, L. Wang, S. Zhang, X. Liu, J. Zhang, Y. Yin, K. Jiao, Q. Du, X. Li, M. D. Guiver, Microstructural orientation of anion exchange membrane through mechanical stretching for improved ion transport, *Front. Membr. Sci. Technol.* 2 (2023), <https://doi.org/10.3389/frmst.2023.1193355>.
- [18] D.R. Dekel, Review of cell performance in anion exchange membrane fuel cells, *J. Power Sources* 375 (2018) 158–169, <https://doi.org/10.1016/j.jpowsour.2017.07.117>.
- [19] G. Merle, M. Wessling, K. Nijmeijer, Anion exchange membranes for alkaline fuel cells: a review, *J. Membr. Sci.* 377 (2011) 1–35, <https://doi.org/10.1016/j.memsci.2011.04.043>.
- [20] F. Xu, Y. Li, J. Ding, B. Lin, Current challenges on the alkaline stability of anion exchange membranes for fuel cells, *ChemElectroChem* 10 (2023), <https://doi.org/10.1002/celec.202300445>.
- [21] J.R. Varcoe, R.C.T. Slade, Prospects for alkaline anion-exchange membranes in low temperature fuel cells, *Fuel Cells* 5 (2005) 187–200, <https://doi.org/10.1002/fuce.200400045>.
- [22] E. Antolini, Lignocellulose, cellulose and lignin as renewable alternative fuels for direct biomass fuel cells, *ChemSusChem* 14 (2021) 189–207, <https://doi.org/10.1002/cssc.202001807>.
- [23] M.A. Hickner, A.M. Herring, E.B. Coughlin, Anion exchange membranes: current status and moving forward, *J. Polym. Sci. Part B: Polym. Phys.* 51 (2013) 1727–1735, <https://doi.org/10.1002/polb.23395>.

- [24] J. Xue, J. Zhang, X. Liu, T. Huang, H. Jiang, Y. Yin, Y. Qin, M.D. Guiver, Toward alkaline-stable anion exchange membranes in fuel cells: cycloaliphatic quaternary ammonium-based anion conductors, *Electrochem. Energy Rev.* 5 (2022) 348–400, <https://doi.org/10.1007/s41918-021-00105-7>.
- [25] M. Hren, M. Božič, D. Fakin, K.S. Kleinschek, S. Gorgieva, Alkaline membrane fuel cells: anion exchange membranes and fuels, *Sustain. Energy Fuels* 5 (2021) 604–637, <https://doi.org/10.1039/d0se01373k>.
- [26] E. Fontanov, F. Galiano, R. Mancuso, D. Talarico, G. Di Profio, L. Guazzelli, C. S. Pomelli, B. Gabriele, A. Figoli, Structure–property relationship of phosphonium- and imidazolium-based polymerized ionic liquids as anion exchange membranes, in: *Euromembrane 2024 (ENER-OL-5)*; <https://Euromembrane2024.Cz/>, 2024.
- [27] M.P.E. Fontanova, R. Ciriminna, D. Talarico, F. Galiano, A. Figoli, G. Di Profio, R. Mancuso, B. Gabriele, G. Angellotti, G. Li, F. Meneguzzo, CytoCell @ PIL: a new citrus nanocellulose-polymeric ionic liquid composite for enhanced anion exchange membrane alkaline water electrolysis, *ChemRxiv* (2024), <https://doi.org/10.26434/chemrxiv-2024-q0xtq>.
- [28] K. Matuszek, S.L. Piper, A. Brzeczek-Szafran, B. Roy, S. Saher, J.M. Pringle, D. R. MacFarlane, Unexpected energy applications of ionic liquids, *Adv. Mater.* 36 (2024) 1–26, <https://doi.org/10.1002/adma.202313023>.
- [29] K. Mishra, N. Devi, S.S. Siwal, Q. Zhang, W.F. Alsanie, F. Scarpa, V.K. Thakur, Ionic liquid-based polymer nanocomposites for sensors, energy, biomedicine, and environmental applications: roadmap to the future, *Advanced Science* 9 (2022) 1–50, <https://doi.org/10.1002/adv.202202187>.
- [30] O. Lebedeva, D. Kulin, L. Kustov, Advanced research and prospects on polymer ionic liquids: trends, potential and application, *Green Chem.* 25 (2023) 9001–9019, <https://doi.org/10.1039/d3gc02131a>.
- [31] M. Watanabe, M.L. Thomas, S. Zhang, K. Ueno, T. Yasuda, K. Dokko, Application of ionic liquids to energy storage and conversion materials and devices, *Chem. Rev.* 117 (2017) 7190–7239, <https://doi.org/10.1021/acs.chemrev.6b00504>.
- [32] L. Manfredy Marquina, L. La Torre Riveros, V. Jauja Ccana, G. Muedas-Taipé, M. Isaacs, A. La Rosa-Toro, Recent studies on polymer electrolytes containing ionic liquids and their applications in lithium-ion batteries, *J. Electroanal. Chem.* 948 (2023), <https://doi.org/10.1016/j.jelechem.2023.117819>.
- [33] T. Zhou, C. Gui, L. Sun, Y. Hu, H. Lyu, Z. Wang, Z. Song, G. Yu, Energy applications of ionic liquids: recent developments and future prospects, *Chem. Rev.* 123 (2023) 12170–12253, <https://doi.org/10.1021/acs.chemrev.3c00391>.
- [34] F. Galiano, R. Mancuso, L. Guazzelli, C.S. Pomelli, J. Bundschuh, J. Rinklebe, S.-L. Wang, C. Apollaro, F. Palumbo, C. Chiappe, A. Figoli, B. Gabriele, Arsenic water decontamination by a bioinspired As-sequestering porous membrane, *Nat. Water* 2 (2024) 350–359.
- [35] K.J.T. Noonan, K.M. Hugar, H.A. Kostalik, E.B. Lobkovsky, H.D. Abruna, G. W. Coates, Phosphonium-functionalized polyethylene: a new class of base-stable alkaline anion exchange membranes, *J. Am. Chem. Soc.* 134 (2012) 18161–18164, <https://doi.org/10.1021/ja3074666>.
- [36] K.J. Fraser, D.R. MacFarlane, Phosphonium-based ionic liquids: an overview, *Aust. J. Chem.* 62 (2009) 309–321, <https://doi.org/10.1071/CH08558>.
- [37] F. Galiano, R. Mancuso, L. Guazzelli, M. Mauri, C. Chiappe, R. Simonutti, A. Brunetti, C.S. Pomelli, G. Barbieri, B. Gabriele, A. Figoli, Phosphonium ionic liquid-polyacrylate copolymer membranes for improved CO₂ separations, *J. Membr. Sci.* 635 (2021) 119479, <https://doi.org/10.1016/j.memsci.2021.119479>.
- [38] M.S.M. Misenan, N.C. Süer, N. Yilmaz Canli, A.S.A. Khair, T. Eren, Synthesis of oxanorbornene-based phosphonium polymeric ionic liquids (PILs) and investigation of their electrical properties, *Mater. Adv.* 5 (2024) 1977–1990, <https://doi.org/10.1039/d3ma00630a>.
- [39] E. Fontanova, R. Ciriminna, D. Talarico, F. Galiano, A. Figoli, G. Di Profio, R. Mancuso, B. Gabriele, G. Angellotti, G. Li, F. Meneguzzo, M. Pagliaro, CytoCell @ PIL: A New Citrus Nanocellulose-Polymeric Ionic Liquid Composite for Enhanced Anion Exchange Membrane Alkaline Water Electrolysis, (n.d.).
- [40] D. Frenkel, B. Smit, D. Frenkel, B. Smit, Chapter 4 – molecular dynamics simulations, in: *Understanding Molecular Simulation*, 2002, pp. 63–107, <https://doi.org/10.1016/B978-012267351-1/50006-7>.
- [41] D.J. Kim, C.H. Park, E. Tocci, S.Y. Nam, Experimental and modeling study of blended membranes for direct methanol fuel cells, *J. Membr. Sci.* 564 (2018) 308–316, <https://doi.org/10.1016/j.memsci.2018.07.016>.
- [42] E. Sireci, G. De Luca, J. Luque Di Salvo, A. Cipollina, G. Micale, Prediction of equilibrium water uptake and ions diffusivities in ion-exchange membranes combining molecular dynamics and analytical models, *J. Membr. Sci.* 668 (2023), <https://doi.org/10.1016/j.memsci.2022.121283>.
- [43] T. Chakraborti, R. Sharma, A.N. Krishnamoorthy, H. Chaudhari, K. Mamtani, J. K. Singh, Unravelling the effect of molecular interactions on macroscale properties in Sustainion anion exchange membrane (AEM) under hydrated conditions using MD simulations, *J. Membr. Sci.* 705 (2024), <https://doi.org/10.1016/j.memsci.2024.122887>.
- [44] T.G. Mason, B.D. Freeman, E.I. Izgorodina, Influencing molecular dynamics simulations of ion-exchange membranes by considering comonomer propagation, *Macromolecules* 56 (2023) 1263–1277, <https://doi.org/10.1021/acs.macromol.2c01743>.
- [45] M. Al Othmi, P. Lin, W. Schertzer, C.M. Colina, R. Ramprasad, J. Sampath, Investigating correlations in hydroxide ion transport in anion exchange membranes from atomistic molecular dynamics simulations, *ACS Appl. Polym. Mater.* (2024), <https://doi.org/10.1021/acsapm.4c01827>.
- [46] H.K. Gallage Dona, T. Olayiwola, L.A. Briceno-Mena, C.G. Arges, R. Kumar, J. A. Romagnoli, Determining ion activity coefficients in ion-exchange membranes with machine learning and molecular dynamics simulations, *Ind. Eng. Chem. Res.* 62 (2023) 9533–9548, <https://doi.org/10.1021/acs.iecr.3c00636>.
- [47] S. Chen, H. Wang, J. Zhang, S. Lu, Y. Xiang, Effect of side chain on the electrochemical performance of poly (ether ether ketone) based anion-exchange membrane: a molecular dynamics study, *J. Membr. Sci.* 605 (2020) 118105, <https://doi.org/10.1016/j.memsci.2020.118105>.
- [48] K. Zhang, W.S. Yu, X. Ge, L. Wu, T. Xu, Molecular dynamics insight into phase separation and transport in anion-exchange membranes: effect of hydrophobicity of backbones, *J. Membr. Sci.* 661 (2022), <https://doi.org/10.1016/j.memsci.2022.120922>.
- [49] C.H. Park, T.H. Kim, D.J. Kim, S.Y. Nam, Molecular dynamics simulation of the functional group effect in hydrocarbon anionic exchange membranes, *Int. J. Hydrogen Energy* 42 (2017) 20895–20903, <https://doi.org/10.1016/j.ijhydene.2017.05.146>.
- [50] J. Huo, W. Qi, H. Zhu, B. Yang, G. He, J. Bao, X. Zhang, X. Yan, L. Gao, N. Zhang, Molecular dynamics simulation on the effect of water uptake on hydrogen bond network for OH[−] conduction in imidazolium-g-PPO membrane, *Int. J. Hydrogen Energy* 44 (2019) 3760–3770, <https://doi.org/10.1016/j.ijhydene.2018.12.090>.
- [51] M. Karibayev, S. Kalybekkyzy, Y. Wang, A. Mentbayeva, Molecular modeling in anion exchange membrane research: a brief review of recent applications, *Molecules* 27 (2022) 1–30, <https://doi.org/10.3390/molecules27113574>.
- [52] S.Y. Sun, X.Y. Nie, J. Huang, J.G. Yu, Molecular simulation of diffusion behavior of counterions within polyelectrolyte membranes used in electrodialysis, *J. Membr. Sci.* 595 (2020) 117528, <https://doi.org/10.1016/j.memsci.2019.117528>.
- [53] 2023 BIOVIA, Dassault Systèmes, Materials Studio, versione 2023, San Diego: Dassault Systèmes, No Title, (n.d.).
- [54] H. Sun, D. Rigby, Polysiloxanes: ab initio force field and structural, conformational and thermophysical properties, *Spectrochim. Acta A Mol. Biomol. Spectrosc.* 53 (1997) 1301–1323, [https://doi.org/10.1016/S1386-1425\(97\)00013-9](https://doi.org/10.1016/S1386-1425(97)00013-9).
- [55] D. Rigby, H. Sun, B.E. Eichinger, Computer simulations of poly(ethylene oxide): force field, PVT diagram and cyclization behaviour, *Polym. Int.* 44 (1997) 311–330, [https://doi.org/10.1002/\(sici\)1097-0126\(199711\)44:3<311::aid-pi880>3.3.co;2-8](https://doi.org/10.1002/(sici)1097-0126(199711)44:3<311::aid-pi880>3.3.co;2-8).
- [56] A. Farahi, A.S. Sadat Kachoei, S. Rowshanzamir, Uncertainty analysis of the molecular dynamics method for simulations of sulfonated poly ether ether ketone proton exchange membrane in fuel cells, *Int. J. Hydrogen Energy* 78 (2024) 123–133, <https://doi.org/10.1016/j.ijhydene.2024.06.281>.
- [57] X. Xu, Y. Cheng, X. Xu, W. Tang, S.S. Liu, C. Mu, Y. Huang, B. Guo, C. Yan, T. Qian, Manipulating the interface relay jumps of OH[−] ad species to accelerate the anode reaction kinetics in direct ammonia fuel cells, *J. Phys. Chem. Lett.* 16 (2025) 3212–3224, <https://doi.org/10.1021/acs.jpclett.5c00384>.
- [58] T.D. Allen MP, Computer Simulation of Liquids, Oxford University Press, 1989.
- [59] T.F. Willems, C.H. Rycroft, M. Kazi, J.C. Meza, M. Haranczyk, Algorithms and tools for high-throughput geometry-based analysis of crystalline porous materials, *Microporous. Mesoporous. Mater.* 149 (2012) 134–141, <https://doi.org/10.1016/j.micromeso.2011.08.020>.
- [60] B. Montgomery Pettitt, M. Karplus, The potential of mean force surface for the alanine dipeptide in aqueous solution: a theoretical approach, *Chem. Phys. Lett.* 121 (1985) 194–201, [https://doi.org/10.1016/0009-2614\(85\)85509-3](https://doi.org/10.1016/0009-2614(85)85509-3).
- [61] B.M. Pettitt, L.X. Dang, A theoretical study of like ion pairs in solution, *J. Phys. Chem.* 94 (1990) 4303–4308.
- [62] M.R. Shirts, J.D. Chodera, Statistically optimal analysis of samples from multiple equilibrium states, *J. Chem. Phys.* 129 (2008), <https://doi.org/10.1063/1.2978177>.
- [63] E. Fontanov, W. Zhang, I. Nicotera, C. Simari, W. van Baak, G. Di Profio, E. Curcio, E. Drioli, Probing membrane and interface properties in concentrated electrolyte solutions, *J. Membr. Sci.* 459 (2014) 177–189, <https://doi.org/10.1016/j.memsci.2014.01.057>.
- [64] A.K. Verma, A.S. Thorat, J.K. Shah, Estimating ionic conductivity of ionic liquids: Nernst–Einstein and Einstein formalisms, *J. Ion. Liq.* 4 (2024) 100089, <https://doi.org/10.1016/j.jil.2024.100089>.
- [65] H. Ju, A.C. Sagle, B.D. Freeman, J.I. Mardel, A.J. Hill, Characterization of sodium chloride and water transport in crosslinked poly(ethylene oxide) hydrogels, *J. Membr. Sci.* 358 (2010) 131–141, <https://doi.org/10.1016/j.memsci.2010.04.035>.
- [66] G.M. Geise, D.R. Paul, B.D. Freeman, Fundamental water and salt transport properties of polymeric materials, *Prog. Polym. Sci.* 39 (2014) 1–42, <https://doi.org/10.1016/j.progpolymsci.2013.07.001>.
- [67] J. Kamcev, D.R. Paul, B.D. Freeman, Effect of fixed charge group concentration on equilibrium ion sorption in ion exchange membranes, *J. Mater. Chem. A* 5 (2017) 4638–4650, <https://doi.org/10.1039/c6ta07954g>.
- [68] N. Yan, R. Sujunani, J. Kamcev, M. Galizia, E.S. Jang, D.R. Paul, B.D. Freeman, Influence of fixed charge concentration and water uptake on ion sorption in AMPS/PEGDA membranes, *J. Membr. Sci.* 644 (2022) 120171, <https://doi.org/10.1016/j.memsci.2021.120171>.
- [69] R.S. Kingsbury, K. Bruning, S. Zhu, S. Flotron, C.T. Miller, O. Coronell, Influence of water uptake, charge, manning parameter, and contact angle on water and salt transport in commercial ion exchange membranes, *Ind. Eng. Chem. Res.* 58 (2019) 18663–18674, <https://doi.org/10.1021/acs.iecr.9b04113>.
- [70] E. Stránská, Relationships between transport and physical–mechanical properties of ion exchange membranes, *Desalin. Water Treat.* 56 (2015) 3220–3227, <https://doi.org/10.1080/19443994.2014.981413>.
- [71] H. Fan, N.Y. Yip, Elucidating conductivity-permselectivity tradeoffs in electrodialysis and reverse electrodialysis by structure-property analysis of ion-

- exchange membranes, *J. Membr. Sci.* 573 (2019) 668–681, <https://doi.org/10.1016/j.memsci.2018.11.045>.
- [72] C. Sequeiros-Borja, B. Surpeta, A.S. Thirunavukarasu, C.J. Dongmo Fomthum, I. Marchlewski, J. Brezovsky, Water will find its way: transport through narrow tunnels in hydrolases, *J. Chem. Inf. Model.* 64 (2024) 6014–6025, <https://doi.org/10.1021/acs.jcim.4c00094>.
- [73] A. Gugliuzza, G. De Luca, E. Tocci, L. De Lorenzo, E. Drioli, Intermolecular interactions as controlling factor for water sorption into polymer membranes, *J. Phys. Chem. B* 111 (2007) 8868–8878, <https://doi.org/10.1021/jp071776q>.